

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2022

Course Code: MEC101

Course: **BASIC MECHANICAL ENGINEERING**

Question Paper No. : UGF011

Max. Marks:100

PART - A

Answer all the questions

10 x 2 = 20 Marks

1. A shaping machine is a reciprocating type of machine tool. Is it possible to create a concave or convex surface using this machine? If possible, write about the operation. If not, state the reason.
2. If someone asks you to suggest appropriate hand forging operations to increase the length of a bar, and decrease the length of another bar. What is your response?
3. Mention the name of the process of converting the rod to wire, and draw its diagram.
4. The core is used in metal casting of some components. Specify the purpose of using core in metal casting.
5. Both arc welding and gas welding are used for joining the parts, each having its own merits and demerits. List any two advantages of the arc welding process.
6. The refrigerants used in the refrigeration process should possess certain safe, economical and thermodynamic properties. Write down any four important properties of refrigerants.

7. The pressure lubrication system consists of an oil sump, oil pump, pressure gauge, and some other parts. State the use of an oil sump, oil pump, and pressure gauge.
8. In I.C engines, both air and water are used as the cooling medium. Specify the applications in which air is used as the cooling medium.
9. Give any four merits of renewable sources of energy.
10. Distinguish between impulse turbine and reaction turbine.

PART – B

Answer all the questions

4 x 15 = 60 Marks

11. The lathe is a versatile machine tool. The cutting tool is mounted on the tool post for a few lathe operations. In some other operations, it is attached to the tail stock. With the help of related sketches, explain any three operations that can be performed by the tool mounted on the tool post and any two operations that can be performed by the tool attached to the tail stock.

(OR)

12. A Rolling mill is used to deform the metallic ingots into various useful sections. Explain the process of rolling in rolling mills. How do you classify the rolling mills? Discuss the working principle of each type.

13. Your friend wants to manufacture a bend pipe made of aluminium. Explain the step-by-step procedure to prepare the green sand mould with suitable sketches.

(OR)

14. Assume that you are going to start a welding shop of your own. In this shop, you are planning to perform oxy-acetylene gas welding. The various equipment is required for doing oxy-acetylene gas welding. List the equipment needed to perform oxy-acetylene gas

welding and mention the purpose of equipment used in oxy-acetylene gas welding.

15. One of your friends has purchased a four-stroke diesel engine car. He wants to know the working of a four-stroke diesel engine. To fulfill his curiosity, explain the working principle of the four-stroke diesel engine with suitable sketches.

(OR)

16. An ignition system is needed for the petrol engines to ignite the air-fuel mixture. The battery ignition system is one of the ignition systems used in the petrol engine. Describe the working of the battery ignition system with the help of a suitable sketch.

17. In a hydroelectric power plant, the potential energy of water is converted into electrical energy using a few integral parts. Draw a schematic layout of a hydroelectric power plant and explain the function of integrated parts of the plant. State the advantages of the hydroelectric power plant over the thermal power plant.

(OR)

18. In an interview, suppose you are asked to explain in detail about the conversion of water into super-heated steam in a high-pressure boiler. To answer that, draw a diagram of any one high-pressure boiler and explain its working in detail.

PART - C

Answer the following

1 x 20 = 20 Marks

19. Assume that you are working as a design and production engineer for a refrigerator manufacturing company. The vapor compression refrigerator is one of the types of refrigerator which is used to reduce and maintain temperature below atmospheric temperature. From the given information, draw an outline sketch showing the arrangement of the various

components of the vapor compression refrigerator and specify the function of each component. (12)

- (b) In this system, the refrigerant is flowing through the pipe which is installed in a refrigerating system. Assume that the pipe installed in the refrigerating system is made by the extrusion process. Explain its process description with the help of a sketch. (8)
